POLARIS

Mentors – Spring Project

Introduction

2026.10.28



THE OHIO STATE UNIVERSITY



Spring Project - Overview

Goal: Give mentees an introduction to what astronomy and physics research looks like.

- Project length: ~ 8 weeks
- Final deliverable: poster & presentation

The focus is on the **learning experience**, not producing novel research results.

Learning Outcomes

- Comfortable discussing background information about their topic to a **conversational** level
- Develop **fundamental research skills** to prepare them for the next step in their scientific journey/critical thinking
  - Basic coding, looking at/analyzing data, making plots, conducting experiments, using scientific instruments
- Get a perspective of learning beyond the structure of traditional coursework

Project Scope

- Takes you ~2hrs to complete
- Expect 8~10 hrs of work for mentee
- Include 1-2 clear results or deliverables for the poster.
- Avoid extensive mathematical or technical prerequisites.

General Advising Advice

- Ask questions about how things are going and how they feel about workload, etc. (try to avoid intense science questions)
- Try to deliver critiques and feedback using the compliment sandwich structure
- Remember that they are first years!
- Be flexible on the timeline; if the project scope needs to change, be accommodating and open to adjusting rather than pushing through
- Be supportive of their work and efforts; affirmation is always good

Double-Paired Mentors

- It’s totally fine to give both your mentees the same project!
- They can work on the project independently, or collaborate and share one poster
  - Ask your mentees if they’re comfortable sharing a project
- If you’re feeling ambitious, you can give them two separate projects (experienced mentors only)

Help we provide

- Project Cookbooks (huge credit to Melanie Angela Zaidel)
- Poster Library
- Project Outline Template & outline feedback
- Human help from M&C

Project Cookbooks

- We have a few projects from former/current mentors available
  - These may serve as inspiration, or complete projects for you to take and use
- We will send out a spreadsheet with project summaries and links to materials
- Let us know if you have a project you want to contribute!

Poster Library

- A giant pdf including all posters from last year
- Use as **inspiration** for designing project, getting an idea of what the deliverable looks like, and mentee final outcomes

Spring Project - Timeline

- End of Fall semester: Write an outline for the project
- Start of Spring semester: Receive feedback for the outline
- Mid-Feburary – Mid-April (8 weeks): Work on project & poster



Advice from Experienced Mentors

Chloe:

- It helps if you could **set up early** in the fall (download software, prepare coding environment, etc)
- I start **talking about different project ideas with mentee** in the fall. Even if we don't end up doing the most fun one, it contributes to nice conversations.
- **Keep reading assignments light**: too much reading could be discouraging (however, it really depends on your mentee's personality!)
- **Build good habits.** Encourage note-taking, reflection, discussion, and organized file management.

Advice from Experienced Mentors

Melanie:

- You will probably **overestimate** your Mentee's programming and reading comprehension skills. This is not meant as an insult to them, just be patient with them and the world they grew up in. Plan to shorten/lengthen the project.

- A little **encouragement** and **affirmation** will go a long way in making your Mentee feel confident and like a scientist. **Tell them that you believe in them** before a challenging part of the project, **then tell them you're proud of them** when they complete a part.

- Don't forget that leading your Mentee through this project will make you a better Mentor and Advisor! Try to have fun!

Small Group Discussion

Direction:
Get in groups of 4~5 people. Try to sit with people you haven't met or worked with before!
Have at least one experienced mentor in the group

At the start of the semester I mentioned I want this course to be beneficial for mentors as well...

There are two goals of this discussion:

1. **Get to know more fellow mentors!** Please try to talk to people you haven't met before, as much as possible

2. **Talk about the project.** Experience mentor share some stories/advice, first-time mentors ask any question on your mind. Everyone share the project ideas you are brewing!

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- **Start with knowing each other**:
  - Your name, what you are interested in / what you work with
  - (optional) Share one fun fact about yourself!
- **Experienced mentors**: What advice would you give? Share your story/experience!
- **First-time mentors**: What question do you have? What are you looking forward to?
- **Everyone**: What potential projects are you thinking about? Why do you want to do this project, and what are you excited about?